

SUPPLEMENTARY MATERIAL

Supplemental Table 1. Associations between daily PM_{2.5} (lagged one day) and mortality in Lima, Peru, 2010-2016. Table reports effects for linear models (per 10 ug/m³ increase in PM_{2.5}) and for quintiles of PM_{2.5} (using quintile 1 as reference). P-values for interaction are tests for interaction between dichotomized temperature (above or below median maximum temperature) and PM_{2.5}.

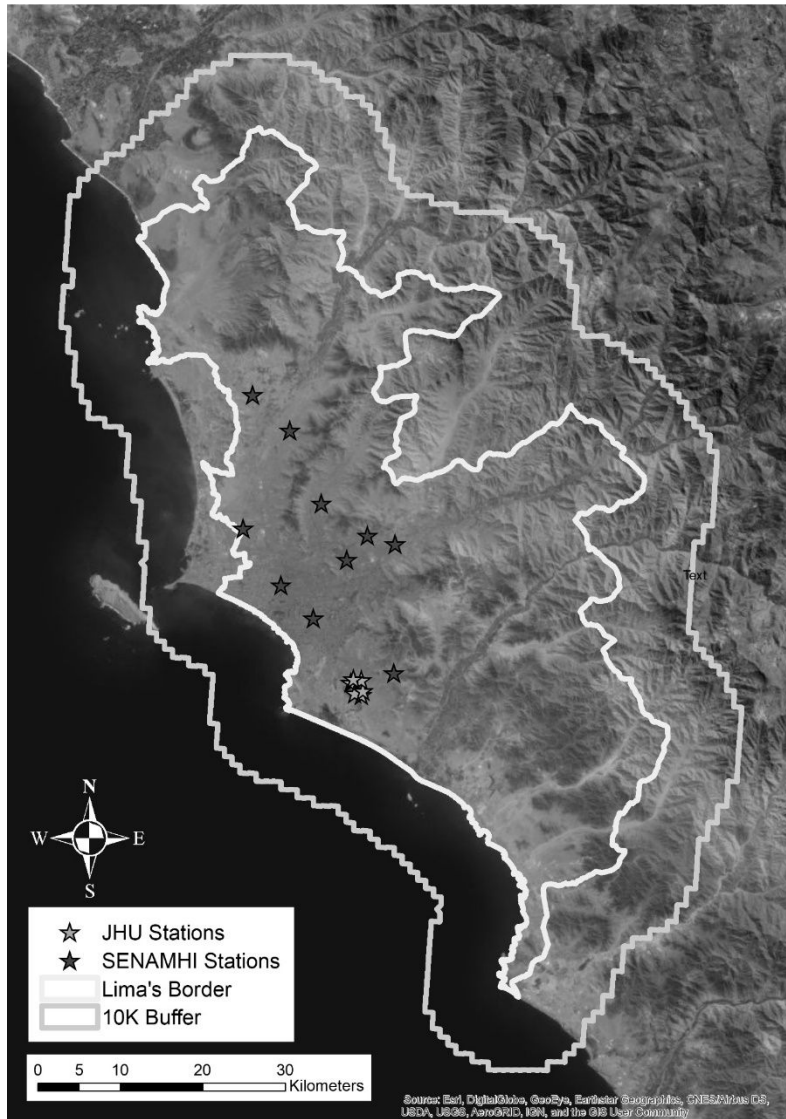
	Rate ratios For mortality	P value for linear trend	P value for interaction
All-causes, linear	1.04 (1.02-1.06)	0.001	0.0002
All-cause , linear, tmax>=med	1.08 (1.04-1.12)	0.0002	
All-causes, linear, tmax<med	1.03 (1.00-1.06)	0.07	
Respiratory causes, linear	1.04 (0.99-1.08)	0.09	0.02
Respiratory causes, linear, tmax>=med	1.11 (1.03-1.19)	0.004	
Respiratory causes, linear, tmax<med	1.04 (0.98-1.10)	0.24	
Circulatory causes, linear	1.07 (1.02-1.13)	0.01	0.002
Circulatory causes, linear, tmax>=med	1.14 (1.05-1.25)	0.002	
Circulatory causes, linear, tmax<med	1.04 (0.98-1.14)	0.21	

Supplemental Table 2. Age-specific temperature-PM_{2.5} interactions for mortality: rate ratios for a 10 ug/m³ increase in PM_{2.5} (linear model).

	Number	Tmax<median	Tmax>=median
All deaths 19-64	41453	0.96 (0.86-1.06)	1.18 (1.04-1.33)
All deaths 65+	144389	1.05 (0.99-1.11)	1.09 (1.01-1.17)
Resp deaths 19-64	11824	1.02 (0.91-1.15)	1.08 (0.94-1.25)
Resp deaths 65+	39791	1.06(0.99-1.14)	1.10 (1.01-1.20)
Circ deaths 19-64	8801	0.97(0.84-1.11)	1.13 (0.96-1.33)
Circ deaths 65+	27742	1.07 (0.99-1.16)	1.10 (0.99-1.22)

Supplemental Table 3. SES-specific temperature-PM_{2.5} interactions for mortality: rate ratios for a 10 ug/m³ increase in PM_{2.5} (linear model)

	Number	Tmax<median	Tmax>=median
All deaths low SES	41453	1.02 (0.98-1.08)	1.03 (0.98-1.08)
All deaths high SES	144389	0.96 (0.91-1.01)	1.06 (0.98-1.14)
Resp deaths low SES	11824	1.06 (0.98-1.14)	1.05 (0.96-1.14)
Resp deaths high SES	39791	0.95 (0.86-1.05)	1.11 (0.96-1.27)
Circ deaths low SES	8801	1.05 (0.96-1.22)	1.09 (0.98-1.22)
Circ deaths high SES	27742	0.98 (0.88-1.10)	1.10 (0.91-1.27)



Supplemental Figure 1. Map of Lima showing the locations of ten automatic $PM^{2.5}$ monitoring stations operated by the Ministry of the Environment (MINAM/SENAMHI) and six $PM^{2.5}$ stations operated by a John's Hopkins University (JHU) research group

27 Supplementary text: Details of spline curves used for Figures 3 and 4.

28 The splines were based on Harrell’s restricted cubic splines (Harrell 1988), using knots at 25%, 50%, and

29 75% of the PM_{2.5} distribution (across all study days), which were 17.46, 19.39, and 23.47 ug/m³,

30 respectively.

31 Hence the spline term was:

32 spline= max(pm-17.46, 0)**3 - ((23.47-17.46)/(23.47-19.39))*max(pm-19.39, 0)**3 + ((19.39-

33 17.46)/(23.47-19.39))*max(pm-23.47, 0)**3

34 where max(pm-17.46, 0) means the maximum of pm-17.46 or 0, whichever is larger.

35 The final model then includes both a linear and spline term for PM_{2.5}.

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